

⇒ chapter 2: Simultaneous Equations

- 1, Substitution Method
- 2, Elimination Method.

$$\rightarrow 2x + 3y = 7$$

$$\rightarrow 3x - 4y = 2$$

sol. by substitution method

from eq 1

$$2x + 3y = 7$$

$$3y = 7 - 2x$$

$$y = \frac{7 - 2x}{3}$$

$$\text{or } 2x = 7 - 3y$$

$$x = \frac{7 - 3y}{2}$$

Now put in sec eq instead of x.

$$3x - 4y = 2 \rightarrow \text{eq 2}$$

$$3\left(\frac{7 - 3y}{2}\right) - 4y = 2$$

$$\frac{21 - 9y}{2} - 4y = 2$$

$$\frac{21 - 9y - 8y}{2} = 2$$

$$21 - 17y = 4$$

$$17y = 21 - 4$$

$$y = \frac{17}{17}$$

$$\boxed{y = 1}$$

Now put this value into any eq.

$$2x + 3y = 7$$

$$2x + 3(1) = 7$$

$$2x + 3 = 7 \quad 2x = 7 - 3$$

$$2x = 4 \quad 2x = 4$$

$$\underline{x = 2} \quad \boxed{x = 2}$$

$$\boxed{x = 3}$$

Prove it:-

$$2(3) + 3(1) = 7$$

$$6 + 3 = 7$$

$$2(2) + 3(1) = 7$$

$$4 + 3 = 7$$